CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of MSc (BT/MI/BC) Examination March 2019

Course code & Name: MS 711 Microbiology

Date: 14/03/2019 Day: Thursday Time: 10:00 am To10:30 am Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 20

1. The following was the first successful solid medium used for isolation of bacteria:
   (i) Agar (ii) Gelatin
   (iii)Potato (iv) Meat
2. The following was the first person to describe microorganisms
   (i) Robert Koch (ii) Antony Van Leewenhoek
   (iii) Louis Pasteur (iv) Sergei Winogradsky
3. The following term describes the exchange of dissolved organic matter (DOM) between microorganisms so that it is recycled many times and not immediately lost to the system.
   (i) Food chain (ii) food web
   (iii) microbial loop (iv) all of the above
4. The major difference between the phase contrast and DIC microscope is the
   (i) light source (ii) path of light
   (iii) λ of light (iv) colour of light
5. According to hydrogen hypothesis the following was the endosymbiont
   (i)anaerobic producing H_2& CO_2 (ii)aerobic utilizing H_2& CO_2
   (iii) Photosynthetic (iv) none of the above
6. Short gaps during comparision between two strands of DNA are observed due to
   (i) gene duplication (ii) unequal crossing over
   (iii) DNA transposition (iv) none of the above

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7. Molecular clock should run faster in organisms with a short generation time is postulated in the following hypothesis
(i) generation- time effect (ii) punctuated equilibria
(iii) Kimura’s model (iv) all of the above

8. Due to the following, sequence dissimilarity is not observed though mutations occur after divergence
(i) parallel substitution (ii) back substitution
(iii) convergent substitution (iv) all of the above

9. PCR amplification **is** required for the following of microorganisms
(i) cultivation (ii) DNA-DNA hybridization
(iii) axenic cultures (iv) G+C content

10. During the reproduction of the following phages the coat protein is assembled when it passes through the cyotplasmic membrane of the host cell
(i) M13 (ii) φX174
(iii) MS2 (iv) T7

11. The following is an example of microcosm:
(i) fermenter (ii) incubator
(iii) Winogradsky column (iv) all of the above

12. The following bacterial sppeciesproduce water by reducing oxygen
(i) *Aquifex* (ii) *Chloroflexus*
(iii) *Hydrogenobacter* (iv) proteobacteria

13 Which of the following is true for blue- green algae?
(i) It is photosynthetic gram -ve bacteria (ii) It cell wall is made up of peptidoglycan
(iii) All blue-green algae pocess phycocyanin (iv) All of the above

14. Saccharomyces cerevisiae belongs to the following .
(i) Ascomycota (ii) Basidomycota
(iii) Chytridia (iv) It is a bacterium

15. What kind of bacteria can be found in lemon pickles?
(i) Thermophilic (ii) Acidophilic
(iii) Alkaliphilic (iv) None of the above

16. Of the following geographical areas where is the likely hood of finding a alkaliphilic bacteria?
(i) Gangotri (ii) Lonar Lake
(iii) Thar desert (iv) All of the above

17 Which of the following culture methods is used for the production of penicillin?.
(i) Batch culture (ii) Continuous culture
(iii) Fed-batch (iv) All of the above

18 Which of the following method is used to enumerate the population size of bacteria?
(i) Petroff-Hausser counting chamber (ii) Viable count
(iii) Measuring the turbidity of the culture (iv) Both b & c

19. The matrix of a biofilm is composed of
(i) Extracellular polymeric substances (EPS) (ii) Flagella
(iii) extracellular DNA (eDNA) (iv) All of the above

20. Which of the following differentiated cell is not present in a biofilm of B.subtilis ?
(i)Miner cells (ii)Matrix producers
(iii) Competent cells (iv) Flagellated cells

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Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I **20**

Q – II A Describe briefly the dispersion strategies of biofilm. **(04)**

OR

Write a short note on different kind of spores found in kingdom Fungi

Q – II B Explain continuous culture and factors affecting the growth in continuous culture **(04)**

OR

Explain the different methods for growing biofilms

Q – II C Using a Petroff-Hausser counting chamber, for a bacterial culture suspension that was stained with PI and Syto9, you are able to count 20 PI positive and 60 Syto9 cells in an area of 1mm2 . Given that the depth of the chamber is 0.1mm. How many live cells are present per ml of the suspension? **(02)**

OR

Describe the different stages in Biofilm formation.

Q – III A **Discuss briefly any two of the following** **(04)**

(i) Genome fusion Hypothesis for the evolution of nucleus

(ii) Core and Pan genome giving suitableexample

(iii) Kimura's two parameter modelfor the study of evolution

Q – III B **Attempt any two of the following** **(06)**

(i) Discuss briefly: the evidence indicating evolution of penicillin resistance in species of *Neisseria* by horizontal gene

(ii) Explain briefly the effect of gene elongation

(iii) List the characteristics of Retropseudogenes derived from Pol II and Pol III transcripts

SECTION - II **30**

Q – IV A **Write a short note on any four of the following** **(12)**

(i) Different phases of growth in batch culture

(ii) Role of surfactin in biofilm formation in *B. subtilis*

(iii) Fed- Batch culture

(iv) Differentiated cells found in cyanobacteria

(v) Water activity and effect of salt on growth of bacteria

Q – V A **Discuss briefly any two of the following** **(06)**

(i) Salient genetic and molecular characteristics of archaea

(ii) Sample preparation for electron microscopy

(iii) Application of Numerical Taxonomy for classification of microorganisms

Q – V B **Write a short note on any one of the following** **(02)**

(i) Microcosm

(ii) Contributions of Louis Pasteur

Q – VI A **Discuss briefly any two of the following** **(06)**

(i) Molecular chronometers

(ii) Nucleic acid hybridization techniques for the study of microbial diversity

(iii) DNA finger printing for the study of microbial diversity

Q – VI B **Attempt any two of the following** **(04)**

(i) List species diversity indices used in study of microbial diversity

(ii) Differentiate between: Physically controlled and biologically controlled ecosystem

(iii) Discuss briefly the application of FISH in study of microbial diversity